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Sleep and Cognition: A Narrative Review Focused on Older Adults

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Abstract

As the number of older adults continues to increase worldwide, an enhanced understanding of age-related changes in sleep and cognition is necessary to slow decline and promote optimal aging. This review synthesizes the extant literature on sleep and cognitive function in healthy older adults, older adults with insomnia, and older adults with sleep apnea, incorporating information on the potential promising effects of treating poor sleep on cognitive outcomes in older adults. Unifying theories of the sleep-cognition association, possible mechanisms of action, and important unanswered questions are identified.

Keywords

Sleep; Insomnia; Sleep apnea; Cognition; Cognitive function; Age; Aging

INTRODUCTION

This article reviews the growing literature examining sleep and cognitive functioning in older adults. The main focus is on normal, age-related cognitive changes, as opposed to neurodegenerative disease processes; however, those are reviewed as appropriate. Age-related cognitive changes are the result of developmental maturation. These cumulative, long-term processes are universal or nearly universal, and are resistant to efforts to reverse

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the change.[1,2] Investigation into cognitive aging has found a general cognitive decline experienced with increasing age,[3–5] which has been shown to be pervasive, affecting many subdomains of cognition, including:

- Reaction time
- Sensory processing
- Attention
- Memory
- Reasoning
- Executive functioning

Although much is known regarding developmental changes in cognitive functioning, comparatively little is known regarding sleep's association with the traditional developmental course of late-life cognitive functioning.

Sleep represents an intriguing individual difference variable because it may relate to late-life cognitive functioning. Sleep has shown consistent age-related changes as a result of developmental maturation. Many of these developmental changes parallel the age-related changes observed in cognitive functioning. For example, slow wave sleep (stage N3) and rapid eye movement (REM) sleep both decrease with advanced age.[6] In addition to these normal, developmentally appropriate changes in sleep, older adults also experience an increased prevalence of both insomnia and sleep disordered breathing (SDB).[7–9] Fig. 1 shows general age-related changes in cognitive functioning and general age-related changes in sleep. The parallel changes in sleep and cognition with age, coupled with anecdotal reports of disturbed cognitive abilities following poor sleep, have resulted in research efforts focused on examining sleep and cognition in older adults. This article summarizes the literature for normal-sleeping older adults, older adults with insomnia, and older adults with SDB.

SLEEP AND COGNITION IN NORMAL-SLEEPING OLDER ADULTS WITHOUT DEMENTIA

Self-Reported Sleep Duration and Cognition

Several large-scale epidemiologic studies have garnered information regarding habitual sleep duration and/or difficulty and cognitive functioning in older adults. In a study of more than 3000 older adults, long sleep duration was associated with worse overall/global cognitive functioning, whereas no association with short sleep duration and cognitive functioning was observed.[10] In a similar study of more than 5000 adults, sleep duration was associated with verbal fluency and list memory, such that both long and short sleep durations were associated with poorer performance.[11] In a sample of community-dwelling older women, sleeping less than 5 hours per night was associated with poorer global cognition and poorer performance across many of the individual indicators of cognitive functioning (i.e., verbal memory, verbal fluency, working memory) compared with women sleeping 7 hours or more per night.[12] The investigators of these studies suggest that sleep duration may be related

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to cognitive functioning through changes in sleep architecture, fragmentation, quality, and neurologic conditions.[10,11]

In additional to individual large epidemiological studies, the association between sleep duration measured via self-reported retrospective recall and cognitive functioning has been corroborated by a meta-analysis showing deleterious effects in both long and short sleepers on multiple domains of cognitive functioning in older adults.[13] Similarly, a review by Devore and colleagues[14] examined cross-sectional and longitudinal studies of sleep duration and cognitive function in older adults. Although actigraphy, polysomnography, and self-reported sleep duration were included in the review, the authors found that all significant associations between sleep duration and cognitive function were from studies using self-reported sleep duration. The review reported 5 of 27 cross-sectional studies and 4 of 7 longitudinal studies found an association between short sleep duration and worse cognitive function, while 9 of 27 cross-sectional studies and 3 of 7 prospective studies found an association between long sleep duration and worse cognitive function. Moreover, shifting to short sleep duration and longer sleep duration from a normal duration of sleep was also associated with poorer cognitive function.

In addition to nocturnal sleep duration, additional cross-sectional and longitudinal evidence suggests sleep duration during the day, or napping, may be associated with cognitive performance in older adults.[15] Older adults who slept longer during the day demonstrated significantly poorer visuospatial ability and slower processing speed than those who slept less during the day. Moreover, sleep during the day was associated with greater decline in visuospatial reasoning and processing speed six years later and remained after controlling for significant health covariates. However, sleep during the day was not significantly associated with verbal memory ability.

Polysomnography-Measured Sleep and Cognition

Investigation into the relationship between polysomnography (PSG)-assessed sleep and waking cognitive functioning has provided mixed results. It has been reported that longer sleep onset latency is related to poorer verbal memory and executive functioning, whereas greater total wake time is related to lower psychomotor speed and memory in normal-sleeping older adults.[13] However, in another investigation, slow wave sleep was unrelated to performance on a simple reaction time task, continuous performance task, and attention test in good-sleeping older adults.[16] Djonlagic and colleagues[17] examined macro and micro sleep architecture and cognitive functioning in two large samples of older adults. Increased REM duration, sleep efficiency, sleep maintenance efficiency, and reduced slow wave oscillation were associated with better cognitive processing speed and executive functioning in older adults. An additional study from Cox and colleagues[15] found no evidence supporting an association between sleep duration and cognitive abilities. These findings are consistent with null findings reported in an earlier systematic review.[14] As such, it seems that additional research is need to further explicate the relationship between PSG-measured sleep and cognitive functioning in older adults without a sleep complaint.

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Actigraphy-Measured Sleep and Cognition

There is some research supporting an association between objective sleep measured with actigraphy and cognitive performance. In a study of nearly 3000 older community-dwelling women, actigraphy-measured sleep (sleep efficiency, sleep onset latency, wake after sleep onset, and napping) was associated with an increased risk of poorer general cognition and executive performance.[18] However, total sleep time was not related to cognitive functioning, which led the investigators to conclude that "it is disturbance of sleep rather than quantity that affects cognition."[18] This conclusion is supported by preliminary evidence showing sleep disruption (the number of wake bouts) is significantly associated with cognitive decline five years later. Furthermore, the association was moderated by sex, with greater wake bouts increasing the odds of cognitive decline for men but decreasing the odds of cognitive decline for women.[19]

In a different but complementary vein, 7 nights of actigraphy were used to compute sleep/wake patterns in 144 community-dwelling older adults. Older adults who displayed many shifts from rest to activity performed worse on composites of executive functioning, memory, and speed than elderly with more consistent rest-activity patterns.[20] Despite some evidence for actigraphy-measured sleep and cognition in late-life, discrepant findings from the systematic review performed by Devore and colleagues[14] suggest there may be additional nuance to this association with older adults. The scarcity of research examining actigraphy-measured sleep and cognition in older adults precludes any definitive conclusions being drawn.

Sleep Deprivation and Cognition

Short-term total sleep deprivation has a significant deleterious effect across most cognitive domains, including attention, working memory, processing speed, short-term memory, and reasoning, with smaller effects being observed for tasks of greater complexity.[21] Webb and Levy[22] and Webb[23] conducted experiments to examine potential age differences in cognitive response to sleep deprivation. In both experiments, older adults' and younger adults' performance on a variety of cognitive tasks were compared following 2 nights of sleep deprivation. Older adults showed greater deterioration following sleep deprivation than did the younger adults in vigilance, visual search, reaction times, word detection, addition, anagrams, and objects uses.[22,23] A similar study was conducted by Pasula and colleagues[24] comparing younger and older adults on visuospatial and verbal working memory abilities following a night of normal sleep and 32 hours of sleep deprivation. Although young adults outperformed older on encoding and displacement process for verbal and visuospatial working memory, older adults showed significantly less decline than younger adults in verbal encoding and visuospatial displacement processes. Following sleep deprivation older adults did not show significant changes in visuospatial working memory, but did demonstrate verbal working memory displacement deficits. Jones and Harrison[25] summarized the extant sleep deprivation work by stating that, "neurocognitive studies present many inconsistencies, task classification is often ambiguous and, in the absence of any unifying explanation at the level of cognitive mechanisms, the overall picture is one of a disparate range of impairment following sleep loss"[25]

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Sleep Restriction and Cognition

The experimental evidence regarding the sleep-cognition relationship garnered through studies using sleep restriction methodology have consistently yielded results indicating an impact on vigilance, which may be reduced in older adults.[26] Similarly, moderation analyses from a meta-analytic review indicated a differential impact of sleep restriction on cognitive function across the lifespan.[27] Although the association between sleep restriction and global cognition function strengthened with age, the association was not significant in older adults. The authors recommended cautious interpretation of the findings due to limited studies in the review examining sleep restriction and cognitive function in older adults. However, they suggested older adults may be more resilient than other age groups to the effects of sleep loss. Relatedly, Bliese and colleagues[26] examined age-related changes in reaction time/attention after modest sleep restriction. Older adults showed less pronounced effects of sleep restriction on their reaction times than younger adults. However, the oldest adult included in the study was 62 years old, so aging effects must be interpreted cautiously. Nevertheless, the investigators suggest that older adults may have "expended more effort across days"[21] resulting in blunted differences.[26] Regarding potential mechanisms underlying the relationship between sleep restriction and cognitive functioning, Banks and Dinges[28] summarized the evidence by suggesting that there is no "definitive evidence of what is accumulating and destabilizing cognitive functions over time when sleep is regularly restricted."[28]

Sleep and Learning

As opposed to examination of the negative consequences of poor sleep (or sleep loss), some researchers have investigated the potential benefits of sleep gained. In this line of research, participants are allowed to sleep while manipulating the timing of cognitive testing/training to either allow sleep to occur following testing or not in order to examine any effects of post training sleep on subsequent testing. Most of this research has been conducted with younger adults with findings indicating optimized performance following sleep. Tucker and colleagues[29] trained 16 healthy older adults in a finger tapping sequencing task and found that older adults performed significantly better following sleep than 12 hours of not sleeping, suggesting sleep-dependent motor skill performance in the elderly. Older adults showed similar rates of improvement as were found in younger samples; however, specific sleep characteristics did not correlate with the next day's performance (in contrast, stage 2 sleep and sleep spindle activity does correlate in younger adults). The investigators concluded that sleep in the elderly does optimize motor skill learning; however, it may do so differently than in younger adults.[29]

SLEEP AND COGNITION IN OLDER ADULTS WITH INSOMNIA

Older adults are at an increased risk for both insomnia and experiencing negative changes in cognitive functioning. Given the comorbidity of insomnia and cognitive dysfunction in older adults, researchers have attempted to understand the role of insomnia in predicting cognitive functioning in a variety of samples (e.g., cognitively intact older adults and older adults experiencing cognitive decline or dementia). Furthermore, a small number of studies

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have examined the effect of behavioral interventions for insomnia on cognitive outcomes. A sample of representative studies is presented later that illustrates these different approaches.

Insomnia Status and Normal Cognitive Aging

The association between insomnia and cognitive performance in younger and middle-aged adults is well-established, with impairments in working memory, episodic memory, and some aspects of executive functioning.[30] However, less is known about the association between insomnia diagnosis and cognitive performance in cognitively healthy older adults who are not experiencing cognitive decline or dementia. To date, insomnia and cognitive functioning have been examined in older adults with insomnia through cross-sectional designs with matched healthy controls[31,32] or using comparisons across insomnia subtypes.[33]

Overall, in contrast with healthy controls, insomnia status was associated with worse performance on a subset of cognitive tasks. Specifically, participants with insomnia performed significantly worse on memory span, integration of visual and semantic dimensions, and executive functioning tasks.[32] The insomnia groups performed better than healthy controls on the simple attending task.[31] The better attendonal performance on simple attending did not persist for complex attending, perhaps reflecting the higher arousal that is characteristic of individuals diagnosed with insomnia.[34] This higher arousal may be beneficial for unambiguous stimuli but a hindrance for complex tasks requiring more cognitive resources.[31] In addition to differences in cognitive performance dependent on the presence or absence of insomnia, performance also seems to differ depending on the type of insomnia complaint. Specifically, Ling and colleagues[33] found that only early morning awakening was associated with significantly worse cognitive performance in the executive functioning domain. No association was found between difficulty initiating sleep or difficulty maintaining sleep across the following cognitive domains: attention, verbal memory, visuospatial ability, and executive functioning. Table 1 lists the differences in cognitive functioning between older adults with and without insomnia.

Overall, across the reviewed studies examining cognitive performance in cognitively intact older adults with insomnia, insomnia status and specific characteristics of insomnia (e.g., early morning awakening) seem to be associated with worse cognitive performance. However, the poorer performance of the individuals with insomnia is not consistent across all cognitive domains or across all tasks within cognitive domains. Consequently, despite a trend toward worse cognitive performance associated with insomnia status, the existing body of evidence is too small and inconsistent to arrive at a broad conclusion.

Insomnia and Cognitive Decline/Dementia Diagnoses

Compared with research that has examined insomnia and cognitive performance in cognitively healthy older adults, more work has been done to explore the role of insomnia in predicting cognitive decline or dementia status. Despite the greater breadth of research with cognitively impaired older adults, the association between insomnia and cognitive decline and/or dementia status remains unclear.

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A minority of studies identified an association between insomnia and cognitive decline or dementia.[35–37] A meta-analysis reported that older adults with insomnia or insomnia complaints had significantly higher risk for incidence of all-cause dementia than those without insomnia at 9 years follow-up.[38] Older adults with long- term insomnia and long-term use of hypnotics had a 2-fold risk of developing dementia during a 3- year follow-up period compared with healthy controls.[35] Controlling for hypnotic use, similar results were found with long-term insomnia predicting an increased risk for cognitive decline at a 3-year follow-up for older adults with insomnia compared with healthy controls.[36] These results point to the importance of considering the duration of the insomnia complaint as a predictor of cognitive function, because long-term/chronic insomnia, rather than concurrent or acute insomnia, has been more consistently associated with cognitive decline.[36]

In contrast, another set of studies found a positive association between insomnia symptoms and cognitive performance.[39,40] Using a longitudinal design, Jaussent and colleagues[39] found that complaints of awakenings during the night and total number of insomnia complaints at baseline predicted a decreased risk for cognitive decline during an 8-year follow-up period. In addition, a cross- sectional approach with older adults with dementia residing in assisted living facilities showed that individuals with insomnia symptoms performed significantly better on the Mini-Mental State Examination (MMSE) than those without insomnia symptoms.[40] Of note, neither of these studies assessed the duration of the insomnia complaint.

A large proportion of studies investigating insomnia and cognitive decline or dementia have failed to report significant associations. Utilizing longitudinal and cross-sectional designs, with and without comparison groups, and various approaches to insomnia and cognitive evaluations, insomnia was not associated with cognitive decline[41–43] or dementia status.[44] Given these discrepant findings, it could be concluded that insomnia status does not predict cognitive decline in older adults. However, it is more likely that the equivocal findings result from methodological differences in research approaches. For example, one study examined the moderating role of insomnia symptoms in the association between elevated amyloid-β and cognitive decline among older adults without dementia.[45] A preliminary analyses revealed that elevated amyloid-β predicted faster cognitive decline in all cognitive domains eight years later, but insomnia was not an independent predictor of cognitive decline. However, older adults with elevated amyloid-β and insomnia exhibited greater rates of decline in memory, executive function, and language compared to those with only elevated amyloid-β or only insomnia or neither.[45] Table 2 provides a summary of the associations between insomnia and cognitive decline/dementia status in older adults.

Insomnia Treatment and Cognitive Functioning

In addition to examining cross-sectional and longitudinal associations between insomnia and cognitive outcomes, a small number of studies have investigated whether the treatment of insomnia leads to an improvement in cognitive performance (Table 3).[31,46] In one such study, an insomnia intervention was used with community-dwelling older adults with insomnia.[31] The insomnia intervention consisted of sleep restriction, cognitive restructuring, sleep hygiene, bright light exposure, body temperature manipulations, and structured

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physical activity. Following treatment, sleep onset latency and sleep efficiency were significantly improved in the treatment group compared with the waitlist group. Treatment was also associated with improved performance on complex vigilance tasks and worsened performance on simple vigilance tasks compared with the waitlist control. A possible explanation for the performance differences on simple versus complex tasks following treatment is that improved sleep results in a reduction of arousal levels to normal,[31] resulting in slower performance on simpler tasks.

Two studies employed a brief behavioral treatment of insomnia approach with a sample of community-dwelling older adults who were cognitively intact and diagnosed with insomnia[46,47] One study found individuals in the treatment group showed a significantly greater decrease in wake time after sleep onset following treatment compared with the information-only control group. However, the treatment group did not show significantly better improvement in cognitive performance across 3 cognitive domains (episodic memory, working memory, abstract reasoning) compared with the control group. The investigators posited that the null findings might be caused by the short follow- up period (4 weeks after the start of the intervention). Allowing more time post intervention for the sleep treatment to take effect might have enabled detection of effects on cognitive performance.[46] Similarly, the second study found significant improvement in actigraphy-measured wake after sleep onset and sleep diary-measured wake time after sleep onset, sleep onset latency, sleep efficiency, and sleep quality. However, no significant improvement was found following the intervention for global cognitive functioning, attention, language, memory or executive functioning.[47]

Building upon this work, McCrae and colleagues[48] examined the patterns of association between nighttime sleep and next-day cognitive performance among older adults following brief behavioral treatment of insomnia.[48] At post-treatment, total sleep time, sleep onset latency, and sleep efficiency were not associated with sustained attention and processing speed. However, greater wake after sleep onset was associated with better next day sustained attention and processing speed. At post-treatment, longer total sleep time was associated with better next-day inductive reasoning performance, but no significant relationships were found between inductive reasoning and sleep onset latency, wake after sleep onset, and sleep efficiency.[48]

SLEEP AND COGNITION IN OLDER ADULTS WITH SLEEP DISORDERED BREATHING

The estimated prevalence of both SDB and cognitive impairment increases with age.[49,50] Moreover, individuals with SDB show cognitive changes similar to those associated with aging.[51] Recent research has posited that SDB and advanced age act independently to impair cognitive functioning, with the combination of both SDB and advanced age leading to cognitive impairments greater than either factor alone.[51]

Although some studies have found an association between SDB and impairments in global cognitive functioning,[52,53] not all cognitive domains seem to be equally affected. Instead, the domains of vigilance, executive function, and memory are particularly implicated.[54]

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The impact of SDB on all 3 of these cognitive domains has been observed among both community-based and clinic-based populations with different neuropsychological tests (Table 4). SDB in older adults has been found to impair:

- Vigilance [55,56]
- Attention[57,58,59,60]
- Reaction time[61]
- Executive functioning[58,62,63,64]
- Problem solving[65]
- Verbal recall[63,66,67]
- Nonverbal recall[65]
- Episodic memory[62]
- Declarative memory[64]
- Processing speed[64]

Cross and colleagues[64] performed a meta-analysis and review of 13 studies examining neuropsychological performance in older adults with obstructive sleep apnea (OSA). Executive functioning was the most common cognitive domain explored in studies. Overall, OSA had a small but significant impact on executive functioning, processing speed, and declarative memory. The effect of OSA was small and non-significant for working memory, global cognition, and motor learning. The meta-analysis did find years of education, mean AHI, and proportion of females in study samples to have a significant moderating effect on the association between OSA and neuropsychological performance. The authors suggest differences in study methodology may have contributed to findings, as small case-control studies from specialized sleep clinics showed significant medium effect sized associations between OSA and cognitive functioning, but larger cohort studies from the community obtained null results.[64] Additional meta-analytic data suggests some sex differences in risk of cognitive decline. Women with OSA are more likely to develop mild cognitive impairment than men with OSA.

Although few studies have examined the longitudinal impact of SDB on cognitive performance in older adults, preliminary evidence suggests that SDB can contribute to relevant long-term changes. A systematic review by Bubu and colleagues[68] found three studies that employed longitudinal designs. Evidence is mixed with two studies reporting positive findings and one reporting null findings. The first found a modest association between nocturnal hypoxemia and global cognitive decline over 3 years among 2636 cognitively normal older adults.[69] The second study found that higher apnea-hypopnea index (AHI) was associated with a slight decline in attention over 8 years for 559 community-dwelling older adults.[70] The final study containing 966 older adults found no association between OSA severity or nocturnal hypoxemia and cognitive decline over 15 years.[71] The authors note that some discrepancy in the findings may due to the methodological differences, including the use of healthy participants with strict inclusion criteria and

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differences between excluded participants and those who remained in the study in terms of AHI and indices and hypoxemia, obesity, hypertension.

The long-term impact of SDB may also influence the onset and course of certain neurologic disorders. Individuals with SDB have higher rates of both mild cognitive impairment and dementia at an earlier age.[72,73] Evidence also suggests that older adults with neurologic disorders might be more vulnerable to the negative cognitive effects of SDB.[56] For example, SDB can exacerbate cognitive impairments in older adults with dementia.[74] The possible role of SDB in the development of neurologic disorders has led researchers to examine the possible pathways contributing to impairments in cognitive functioning. According to one review, evidence suggest a link between OSA and biomarkers of Alzheimer's disease pathology.[68] While longitudinal evidence is still limited, findings suggest OSA severity is associated with higher amyloid burden over a 2-year period. An additional longitudinal analysis that included cognitive normal older adults, as well those with various dementias, and mild cognitive impairment, self-reported diagnosis of OSA was associated with larger increases in amyloid burden and cerebral spinal fluid concentration over 2.5 years in cognitively normal older adults and older adults with mild cognitive impairment, but not older adults with Alzheimer's disease.[68]

Several mechanisms through which SDB contributes to cognitive decline in older adults have been proposed. Evidence suggests that high levels of hypoxemia, an abnormally low level of oxygen in the blood, may play a prominent role. Yaffe and colleagues[73] found that hypoxemia was associated with an increased risk of future mild cognitive impairment and dementia. In addition, increased hypoxemia is associated with greater impairment in global cognitive functioning, as well as declines in specific cognitive domains. [55,75] However, other studies have found no association between hypoxemia and impaired cognition.[52,67] As a result, both sleep fragmentation and excessive daytime sleepiness have been proposed as alternative mechanisms through which SDB may lead to cognitive dysfunction.[76] For example, excessive daytime sleepiness caused by SDB was associated with a decline in global cognitive functioning.[52] Despite these findings, the relative contributions of both sleep fragmentation and excessive daytime sleepiness to cognitive impairment remain poorly understood. Interestingly, SDB may manifest differently in men and women, with men displaying more episodic hypoxia and women having more sleep fragmentation. These sex-specific presentations of SDB may be associated with unique cognitive sequela.

Although strong evidence supports an association between SDB and cognitive decline in older adults, not all findings support this association. One study found that although both SDB and aging are independently associated with cognitive deficits, age did not interact with SDB to make those cognitive deficits worse.[77] Other studies report similar findings. Boland and colleagues[78] found no evidence for an association between either mild or moderate forms of SDB and cognitive functioning in verbal learning and short-term recall, psychomotor efficiency, or verbal fluency. Similarly, other research found no association between SDB and performance on broad standardized cognitive tests.[79,80] Studies that report no association between SDB and cognitive impairment in older adults contain several methodological differences. First, they generally lacked comprehensive measures of cognitive functioning or relied on single measures of global functioning [78] Cognitive

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impairments in older adults with SDB may be subtle and may not be readily identified by general/global cognitive measures. Second, several of the studies that report no association between SDB and cognition used home-based PSG,[80] which has been shown to differ from laboratory PSG, particularly for adults with severe SDB.[81] In addition, several studies reporting no associations between SDB and cognitive performance only included participants with mild to moderate SDB,[78,79] potentially suggestive of a dose-response association between SDB severity and cognitive performance.

Although the findings are inconsistent, overall, current research suggests that SDB adversely affects cognitive performance in older adults. Cognitive impairments are particularly pronounced in the domains of attention, executive function, and memory; however, these impairments may be sex-specific. These findings are largely consistent with research examining the influence of SDB in young and middle-aged adults.[82,83]

Sleep Disordered Breathing Severity and Cognitive Performance

Research has explored whether the severity of SDB (as indexed by the apnea-hypopnea index [AHI] or respiratory disturbance index [RDI]) is related to the level of cognitive impairment in older adults. Studies have found that greater AHI and RDI were associated with poorer global cognitive functioning,[53] as well as adverse outcomes on specific cognitive domains such as:

- Vigilance[56,59,92]
- Executive functioning[63]
- Language functioning[93]
- Attention[58,60]
- Executive function[58,60]
- Memory[66]

However, one study examining older adults with mostly mild to moderate SDB found no association between SDB severity and cognitive functioning,[78] suggesting that there may be a threshold at which SDB exerts an increasingly negative impact on cognition. Further research investigating SDB severity and cognitive impairment is warranted.

Sleep Disordered Breathing Treatment and Cognitive Performance

Given the nature of cognitive dysfunction in older adults with SDB, as well as the association between SDB and neurologic disorders, the influence of SDB treatment on cognitive functioning in older adults has been the subject of recent interest. The most common treatment of SDB is positive airway pressure (PAP) therapy. Although PAP has been shown to decrease some of the primary sequelae of SDB, such as sleep fragmentation and nocturnal oxygen saturation, several studies suggest possible additional benefits of PAP treatment on cognitive performance.

Four studies report improvements in many areas of cognitive functioning following 3 months of PAP treatment.[66,94,95,96]

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Improvements were shown in the following domains:

- Episodic learning and memory
- Short-term memory
- Executive functioning
- Working memory
- Attention
- Psychomotor speed
- Nonverbal delayed recall

Despite optimism about the potential benefits of PAP on cognition, some studies have shown more limited and uneven benefits of PAP treatment. For example, Kang and colleagues[97] found that short-term PAP treatment was associated with gains in executive functioning but no other cognitive domains. Moreover, small single-center trials have shown improvement in attention, psychomotor speed, memory, and executive function while larger, multicenter clinical cohorts show significant improvement in only working memory or no significant changes.[68] Low adherence and short-term PAP usage were suspected have weakened the impact of the intervention. Additionally, authors considered a ceiling effect due to high baseline cognitive scores among participants.

Preliminary evidence suggests that cognitive gains observed with short-term PAP treatment might be maintained over time. PAP treatment over the course of 10 years is associated with better memory, attention, and executive functioning.[62] The long-term benefits of PAP treatment may also extend to older adults with neurologic disorders. Preliminary examination of long-term PAP use among individuals with Alzheimer's disease found that treatment can slow cognitive deterioration.[98] Another recent study reported that PAP treatment may delay the age of onset of mild cognitive impairment.[72]

Although the influence of PAP treatment on cognitive outcomes seems promising, noncompliance with the treatment remains prevalent in the general population, as well as among older adults.[99] Limited compliance may severely limit the benefits of PAP treatment on cognitive abilities. Older adults who complied with treatment (average use of 8.5 hours per night) showed greater cognitive abilities compared with individuals who were noncompliant (3.9 hours per night). [90] One hypothesis about PAP compliance is that older adults who notice cognitive gains in readily observable domains, such as attention and memory, may be more likely to comply with PAP treatment.[90]

Although preliminary evidence shows improved cognitive outcomes for older adults treated with PAP, further research is needed to confirm these findings and better understand which cognitive domains are affected. In addition, future research should examine whether other forms of SDB treatment, such as weight loss, oral appliance therapy, positional therapy, and surgical treatments, might lead to similar cognitive gains.

Hypoglossal nerve stimulation (HNS) is a novel, promising treatment for SDB. The treatment utilizes a surgically-implanted device to produce electrical stimulation of the

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hypoglossal nerve in order to move the tongue and relieve upper airway obstruction. Two systematic reviews have shown that HNS treatment significantly improves sleep apnea severity and excessive sleepiness in adult populations at 1- and 5-year follow-up. [100,101] Additional evidence is needed to determine the impact of HNS treatment on cognitive functioning among older adults. This is particularly important given recent evidence suggesting that middle-aged and older adults are among those interested in obtaining HNS treatment[102]

UNIFYING THEORIES AND MECHANISMS

There are several informative hypotheses concerning the association between sleep and cognitive functioning. These hypotheses include the controlled attention hypothesis,[109] neuropsychological hypothesis,[21,25,103] vigilance/arousal hypothesis,[104,105] and wake-state instability hypothesis.[106–108] These various theories about the role of sleep processes in regulating and maintaining cognitive functioning may not be mutually exclusive.[21] The controlled attention and vigilance/arousal hypotheses are essentially parallel descriptions of the same phenomena. Many higher order cognitive functions may rely on the appropriate levels of attention and arousal. It has been suggested that impairment of the prefrontal cortex may cause decrements in attention and vigilance.[110] Sleep seems to impede arousal/vigilance/attention and prefrontal functioning, potentially through instability of the neurobiological systems responsible for attentional and sleep drives. Table 5 provides a description of each sleep-cognition theory. In addition to the sleep-cognition hypotheses discussed earlier, there are many suspected mechanisms involved in the sleep-cognition relationship in late life. These mechanisms are graphically depicted in Fig. 2.

SUMMARY AND FUTURE DIRECTIONS

Sleep and cognitive functioning are linked in late-life; however, the exact nature of this relationship has yet to be discerned. Future studies should continue to investigate the gamut of sleep-cognition relationships. Important questions remain concerning (1) the role of normal sleep changes in normal cognitive aging; (2) the role of pathologic sleep changes in the development of dementias; (3) the utility of treating sleep disorders for improving cognitive functioning, warding off unwanted cognitive decline, and slowing the course of neurodegenerative diseases; and (4) sex-differences in the association between sleep (healthy and disordered) and cognitive functioning. An intriguing prospect for future study is to examine the additive impact of treating sleep disorders in conjunction with focused cognitive interventions, as-well-as treating sleep disorders in older adults with comorbid conditions, like diabetes, known to be associated with cognitive functioning. Perhaps the combination of interventions that focus on different pathways of change may have a synergistic effect and result in more pronounced cognitive improvements. Increasing knowledge of the ways in which sleep may affect late-life cognitive functioning could have far-reaching benefits.

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KEY POINTS

- Sleep and cognitive functioning both show negative changes with advanced age.
- Although the associations are varied, sleep seems to be related to cognitive functioning within good-sleeping older adults, older adults with insomnia, and older adults with sleep disordered breathing.
- Both insomnia and sleep apnea may be associated with cognitive decline and dementia.
- Treatment of sleep disorders may provide cognitive benefits in late life. Additional research is warranted.
- Examination of sex-differences in the sleep—cognition association may address unanswered questions.

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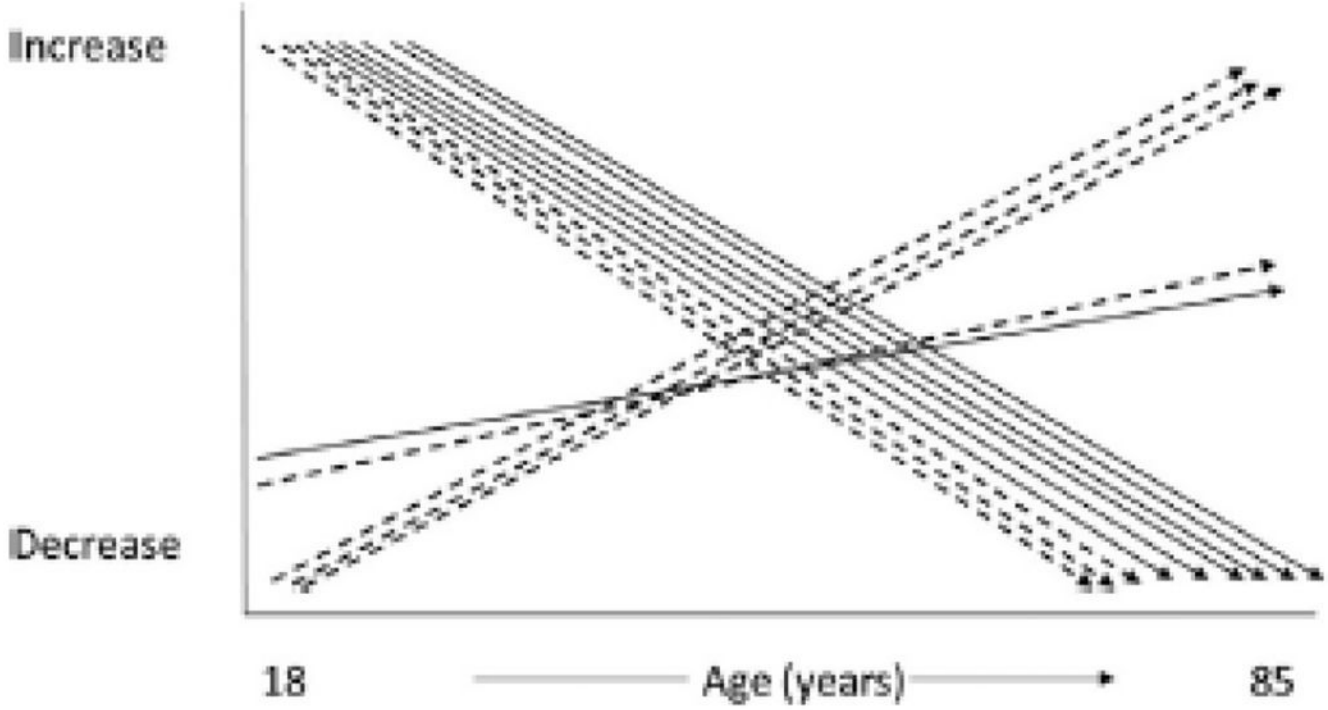

Fig. 1.

Normative changes with age in both cognitive functioning and sleep. Solid arrows represent general cognitive changes. Dashed arrows represent general sleep changes. Cognitive abilities that decline with age include processing speed, working memory, long-term memory, attention, reasoning, and executive control. Sleep characteristics that decline with age include total sleep time, slow wave sleep, and REM sleep. Sleep characteristics that increase with age include wake time after sleep onset, and light sleep (stages N1 and N2). Crystalized intelligence and sleep onset latency both showed slight increases with advancing age.

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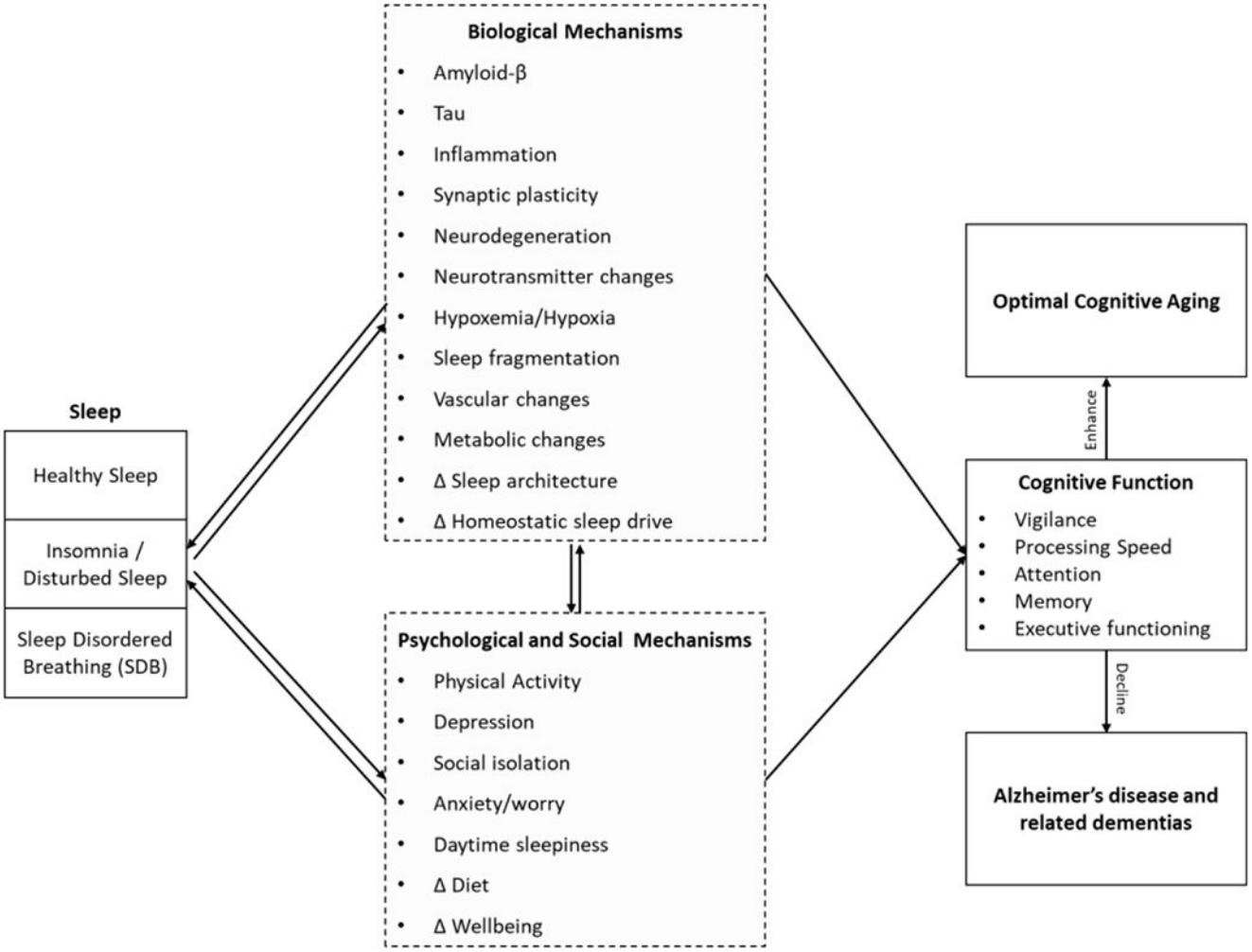

Fig. 2.

Model depicting the relationships among healthy sleep, insomnia/disturbed sleep, and sleep disordered breathing and cognitive functioning, optimal cognitive aging, and Alzheimer's disease and related dementias in older adults. Factors listed in dashed boxes represent potential biopsychosocial mechanisms through which sleep may impact cognitive functioning.

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Table 1

Differences in performance across cognitive tasks group by insomnia status

Cognitive Task	Better Performance by Insomnia Group	Better Performance by Healthy Control Group	No Group Difference
Sustained attention	X (simple tasks)[31]	X (simple tasks,[32] complex tasks[31])	X (complex tasks)[32]
Naming	—	—	X[32]
Psychomotor skills	—	—	X[32]
Memory span	—	X[32]	—
Integration of 2 dimensions (visual and semantic)	—	X[32]	—
Time estimation	—	X[32]	—
Executive functioning	—	X[32]	—

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Table 2

Insomnia status and cognitive decline or dementia diagnosis

Insomnia Measure	Cognitive Measure	Results	Study
Negative Associations			
ICD-9 codes	ICD-9 codes	Patients with insomnia diagnosis and prescribed hypnotics, > risk for dementia during the 3-y follow-up compared with controls	35
Interview indicating either problem most of the time: trouble falling asleep or waking up too early and not falling asleep again	Pfeiffer's SPMSQ, >2 errors on the SPMSQ classified as cognitive decline	Insomnia associated with increased risk of cognitive decline for men, independent of depression and comorbid with depression. Insomnia associated with increased risk of cognitive decline for women when insomnia was comorbid with depression	36
Positive Associations			
Interview and questionnaire: difficulty initiating sleep, awakenings during the night, early morning awakening, insomnia severity	Incident cognitive impairment defined as 4-point reduction in MMSE, 4-point reduction in BVRT, and 14-point reduction in the IST scores	Number of insomnia complaints and difficulty with awakenings during the night were negatively associated with MMSE cognitive decline during follow-up. No associations found for BVRT or IST	39
Johns Hopkins Alzheimer's Disease Research Center questionnaire	Severity of dementia was classified based on MMSE scores (mild = MMSE >20; moderate = MMSE 11-19; severe = MMSE <10)	Participants diagnosed with insomnia performed better on the MMSE than those without sleep disturbance or those with insomnia and daytime sleepiness	40
Adapted Brief Insomnia Questionnaire (BIQ)	A composite score was produced based on a global test that included several cognitive domains. Scores ranges from 0-35. 0-6: Dementia 7-11: cognitively impaired with no dementia (MCI) 12-27: not cognitively impaired	Time-varying insomnia symptoms associated with MCI and dementia after controlling for covariates including age, race, gender, education, BMI, smoking status, drinking, and chronic disease.[37]	37
No Associations			
Self-report questionnaire assessing "Usually having trouble falling asleep or waking up far too early and not going back to sleep"	CASI; cognitive decline was defined as a >9-point decrease in the CASI	Insomnia not associated with greater risk for dementia diagnosis or cognitive decline	41
Interview assessing difficulty falling asleep, staying asleep, or both	Incident cognitive impairment defined as an MMSE <21	No sleep problems associated with increased risk for incident cognitive impairment at 2 and 10 year follow-ups	42
Clinical interview: at least 1 symptom (difficulty falling asleep, staying asleep, early morning awakening) occurring at least 3 times a week	MMSE; Global Deterioration Scale. Participants with MMSE <24.0 were evaluated for dementia by expert panel	Insomnia diagnosis was not associated with cognitive impairment	43
Insomnia Interview Schedule and Sleep Impairment Index: onset, maintenance, termination insomnia	Alzheimer's disease diagnosis was made according to NINCDS/ADRDA and DSM-IV criteria	Individuals with Alzheimer's disease did not differ from the healthy comparison group in terms of the frequency of onset, maintenance, termination of insomnia symptoms	44

Abbreviations: ADRDA, Alzheimer's Disease and Related Disorders Association; BVRT, Benton Visual Retention Test; CASI, cognitive abilities screening instrument; DSM-IV, diagnostic and statistical manual of mental disorders, fourth edition; ICD-9, International Classification of Diseases, Ninth Revision; IST, Isaacs Set Test; MMSE, Mini-Mental State Examination; NINCDS, National Institute of Neurological Disorders and Stroke; SPMSQ, short portable mental status questionnaire.

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Table 3

Insomnia treatment and cognitive performance

Insomnia Measure	Cognitive Measure	Results	Study
DSM-IV criteria, Pittsburgh Sleep Quality Index, Sleep Disorders Questionnaire. PSG and sleep diaries were also used	Simple and complex sustained attention assessed via computer. Outcomes: lapses, false- positive responses, and reaction times	Compared with the waitlist control, insomnia intervention group showed increased reaction time for simple vigilance task and reduced reaction time for complex vigilance task	31
DSM-IV-TR, ICSD-2 verified by self-report questionnaires and clinical interview. Polysomnography and the Pittsburgh Sleep Quality Index were also used	Neuropsychological tests assessed 3 cognitive domains: episodic memory, working memory, abstract reasoning	Insomnia intervention was not associated with greater improved cognitive performance compared with information-only control	46
DSM-5 criteria, verified by clinical interview. Single night ambulatory monitoring for SDB.	Overall cognitive functioning, attention, processing speed, language, memory, executive functioning	No significant improvement in overall cognitive functioning or cognitive domains from pre- to post-treatment.	47

Abbreviations: DSM-IV-TR, DSM-IV text revision; ICSD-2, International Classification of Sleep Disorders, second edition.

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Table 4

Cognitive performance in older adults with sleep disordered breathing

			Representative
Cognitive Domain	**Measures**	**Outcomes**	**References**
Global functioning	• MMSE	• EDS is associated with global cognitive decline over 3 years	84,85
		• Increased levels of hypoxemia and AHI are associated with lower global cognition functioning	
	• Modified Mini-Mental State Exam (3MS)	Nocturnal hypoxemia associated with global cognitive decline over 3 years	69
Vigilance	• Digital Vigilance Test	• SDB is associated with decreased vigilance	55,86,87,88
	• Psychomotor Vigilance Test	• Increased hypoxemia is associated with decreased vigilance	
	• Trail-making Test Part A	• Increased RDI is associated with decreased attention	
	• Digit Symbol Substitution/Coding Subtest	• Older adults with SDB have slower reaction times compared with age-match controls and younger adults	
	• Vienna Test System		
Executive function	• Stroop Color and Word Test	• Higher RDI associated with decreased executive function	58,87,89,63
	• Trail-making Test Part B	• RDI associated with lower cerebral efficiency	
	• Similarities Subtest	• SDB is associated with a greater decline of executive function over time	
	• Raven Progressive Matrices	• Severe SDB is associated with poorer executive functioning	
		• SDB is associated with poorer problem-solving abilities	
Memory	• Hopkins Verbal Learning Test	• SDB is associated with lower nonverbal delayed recall	65,90,91
	• Brief Visuospatial Memory Test-Revised	• Increased AHI and RDI are associated with decreased verbal delayed recall memory	
	• Rey Auditory Verbal Learning Test	• Severe SDB is associated with poorer episodic memory over time	
Attention	• Trail Making Test Part A • Stroop Color-Word Test Part I and II • Coding subtest of WAIS-III	AHI was associated with a small decline in attention over 8 years	70

Abbreviations: AHI, apnea-hypopnea index; EDS, excessive daytime sleepiness; RDI, respiratory disturbance index.

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Table 5

Theories on the link between sleep and cognition

Name	Description	Reference
Controlled attention	Monotonous tasks are most affected by sleep loss because of the amount of top-down control needed to sustain attention, whereas more complex/difficult tasks are intrinsically motivating (i.e., bottom-up control)	39
Neuropsychological	Sleep loss results in focal impairment in functions subserved by the prefrontal cortex (i.e., executive functions), beyond any impairment in attention or vigilance	21,25,103
Vigilance/arousal	Attention, which is needed for the performance of many other cognitive tasks, is mediated by arousal; a common correlated feature of disturbed sleep	104,105
Wake-state instability	Cognitive deficits observed as a result of sleep loss occur because of the interaction of the drive to maintain alertness and the homeostatic drive to initiate sleep	106–108

Please note that these theories are not mutually exclusive but attempt to explain similar phenomena in different ways.